ACADEMIC YEAR: 2025-26

Course: Skill base Lab Course: Cloud Computing

Course code: CSL605

Year/Sem: TE/VI

Experiment No.: 08
Aim: To study and implement Identity and Access Management (IAM) practices on AWS
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Roll Number: 40
Date of Performance: 09/03/2026
Date of Submission: 11/03/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	2
Understanding	4-5	2-3	2
Journal work and timely submission.	8-10	5-8	4

Checked by

Name of Faculty : Dr. Tatwadarshi Nagarhalli

Signature :

Date :

Experiment No. 8

Aim: To study and implement Identity and Access Management (IAM) practices on AWS/Azure cloud

Objective: Understand the working of Identity and Access Management IAM in cloud computing and to demonstrate the case study based on Identity and Access Management (IAM) on AWS/Azure cloud platform

Theory:

- Identity Management is a set of business processes, and a supporting infrastructure, for the creation, maintenance and use of digital identities.
- IAM is an essential function for protecting the privacy of information, enhancing user experience, enabling accountability, and controlling access to an organization's assets.
- IAM is the collection of processes and technology used to manage digital identities and the resource access provided through them.
- Components of access management
    - Establishing unique identities and associated authentication credentials.
    - Authoritative source is maintained as a central repository for storage.
    - Providing capability to identities to request entitlements
    - Assigning roles or entitlements to identities.
    - Managing off boarding and other business work processes by workflows
    - Providing capability to approve, revoke, review or certify entitlements or roles assigned to users.

Steps:

----- Configuring IAM Dashboard -----

1. Go to IAM dashboard
2. Click on create option under Account Alias and give a valid name; save changes
3. (Download Google Authenticator from PlayStore in your Mobile Phone)

----- Configuring IAM Dashboard -----

1. Click on "users" in the left column
2. Click on Add users button
3. Set a custom valid psw (Imc: Qwertyuiop123) and check the Require psw rest box which will make you create a next psw in the next sign in
4. Click on Next: Tags
5. Add a tag if you want to just to keep track of your activities; then click on Next: Review

6. Click on Create User Button
7. Open the URL in Incognito Mode (Imc: https://nimitjjw.signin.aws.amazon.com/console) Open the URL in Incognito Mode (Imc: https://nimitjjw.signin.aws.amazon.com/console)

----- Logging in as the new User & Checking their permissions -----

1. Enter the new user's name and psw saved earlier
2. Enter a new valid psw
3. After logging in, you will notice that you don't have permission to do anything yet

----- Adding MFA for the user via Root User -----

1. Type "AWS CLI" in a new window of any browser and go to it's the main page of AWS regarding the same Click on 64-bit hyperlink in the RHS column under the Windows section and download, install the AWS CL
2. Type "cmd" in the windows search bar and run it as an administrator
3. Type aws configure, it will ask for a few inputs; AWS Access Key ID and Key are the ones which we saved earlier Default region name is whichever region AWS you are using; in case of Mumbai, its: apsouth-1 The output format is json in our case
4. The next two steps are
   OPTIONAL: aws –version
   aws s3 ls

0. Go in the security credentials tab under Users of IAM Dashboard
0. Click on the "Manage" Hyperlink
0. Use the Google Authenticator app downloaded earlier to scan the QR Code
0. Enter two of the codes which are shown in the Google Authenticator App over a span of 30 secs each; click on Assign MFA Button

----- Logging in as the new user after MFA -----

1. Again try logging in via the new user created earlier; this time it will ask for MFA after you click on Sign In
2. Use the code being shown in the Google Authenticator
3. Now, after opening the root user window again After going in the Users section of IAM Dashboard, we can see that MFA has been activated for the new user

----- Adding 3 More Users and Giving them permissions ----

1. Now, Adding 3 More Users
2. Not giving them an Access key and not checking the Psw Reset Checkbox; Click on the Next: Permissions
3. We will create a group later We can see the previous user listed under the copy "permission from existing user" section (just for observation purpose) Click on the third section: Attach existing policies directly
4. Type in ec2fullaccess in the search box and click the check box for it; click on Next: Tags
5. Input the Key and Value for the Tag to keep track of your activities; Click on Next: Review

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6. Click on Create Users Button

----- Logging in as one of the 3 new Users and Checking their permissions ----

1. Try logging in as one of the 3 new users just created
2. Try launching an EC2 instance via the new user
3. Hence, an instance has been created
4. Delete the bucket when done with your work

----- Creating a new Group and Giving it permissions ----

1. Select the members to be present in the group (max 4 per group)
2. Giving this group ec2fullaccess and s3fullaccess

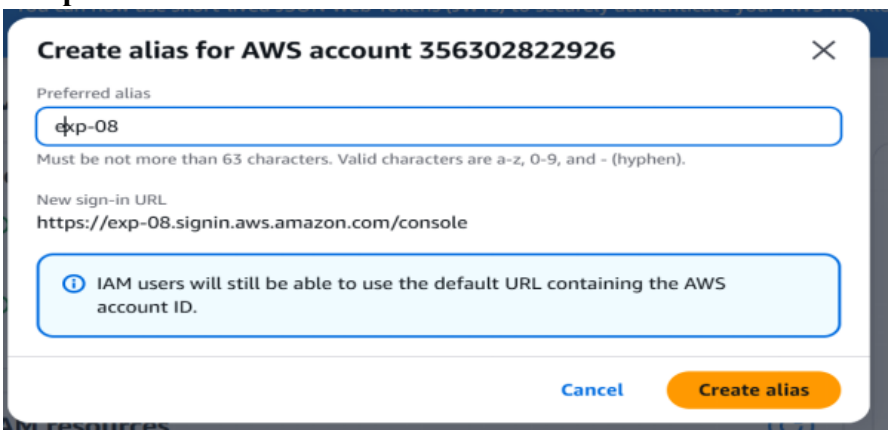
- Logging in as a member of the Group & Checking their permissions –

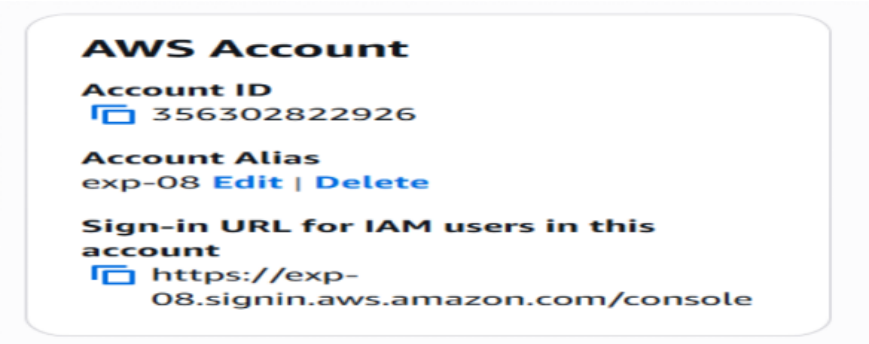
1. Now, login as one of the users from the group and try creating a S3 bucket
2. S3 bucket successfully created
3. Delete the bucket when done with your work

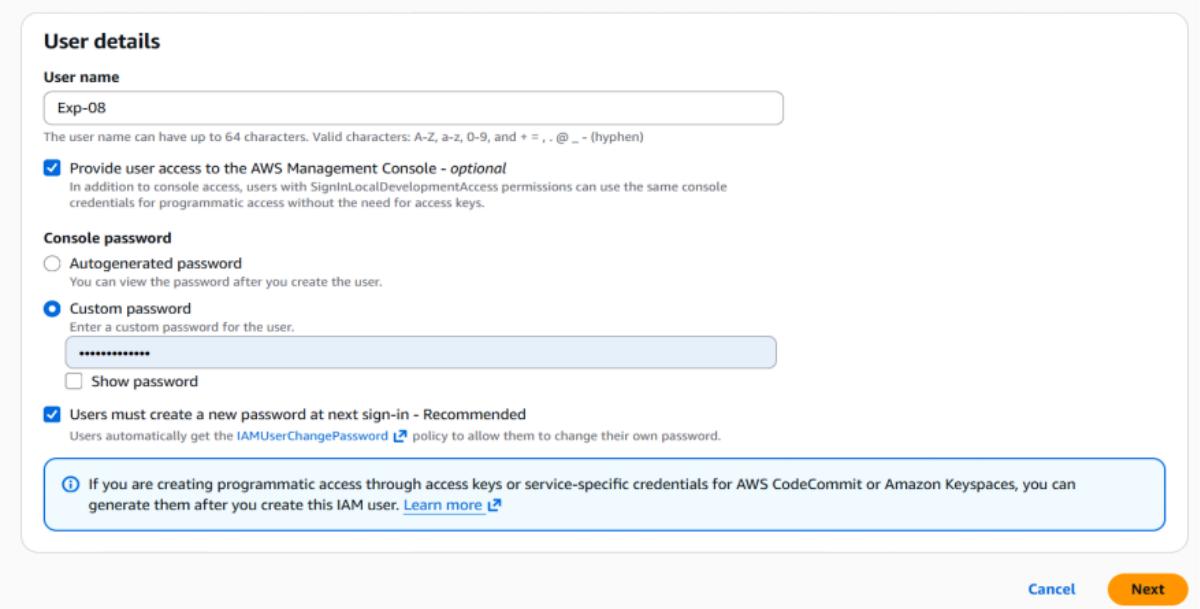
----- Creating a new Role -----

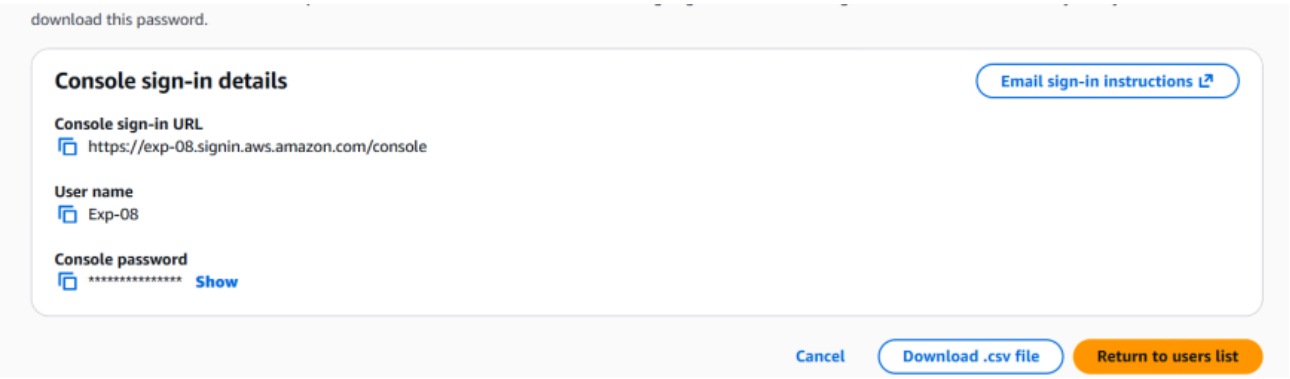
1. Go in the root user window and click on "create role" button in the "Roles" section of IAM Dashboard
2. Let it be the default options (you can choose any use case you like) Click in Next button
3. Give the permission suitable to the use case chosen
4. Give suitable Role name and description; rest would remain as default
5. Add a tag if you want to; click on Create Role button
6. The role has been successfully created
7. Just to check the overall users, groups and roles, you can check out the IAM Dashboard

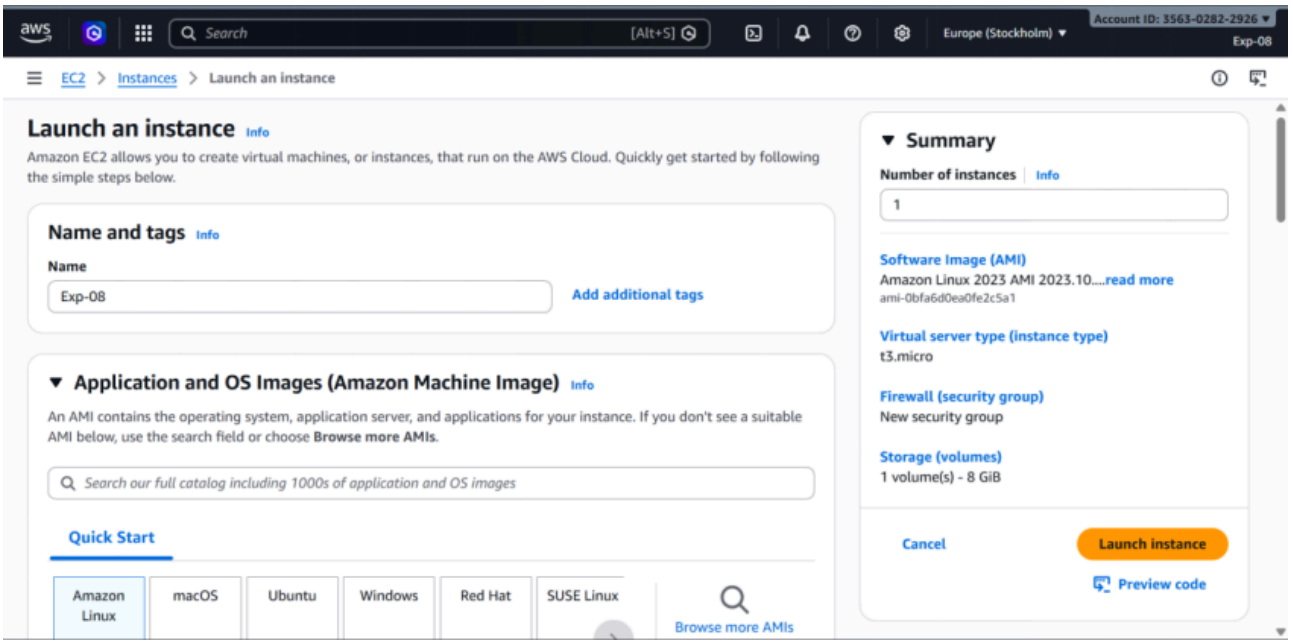
Output/Observation:

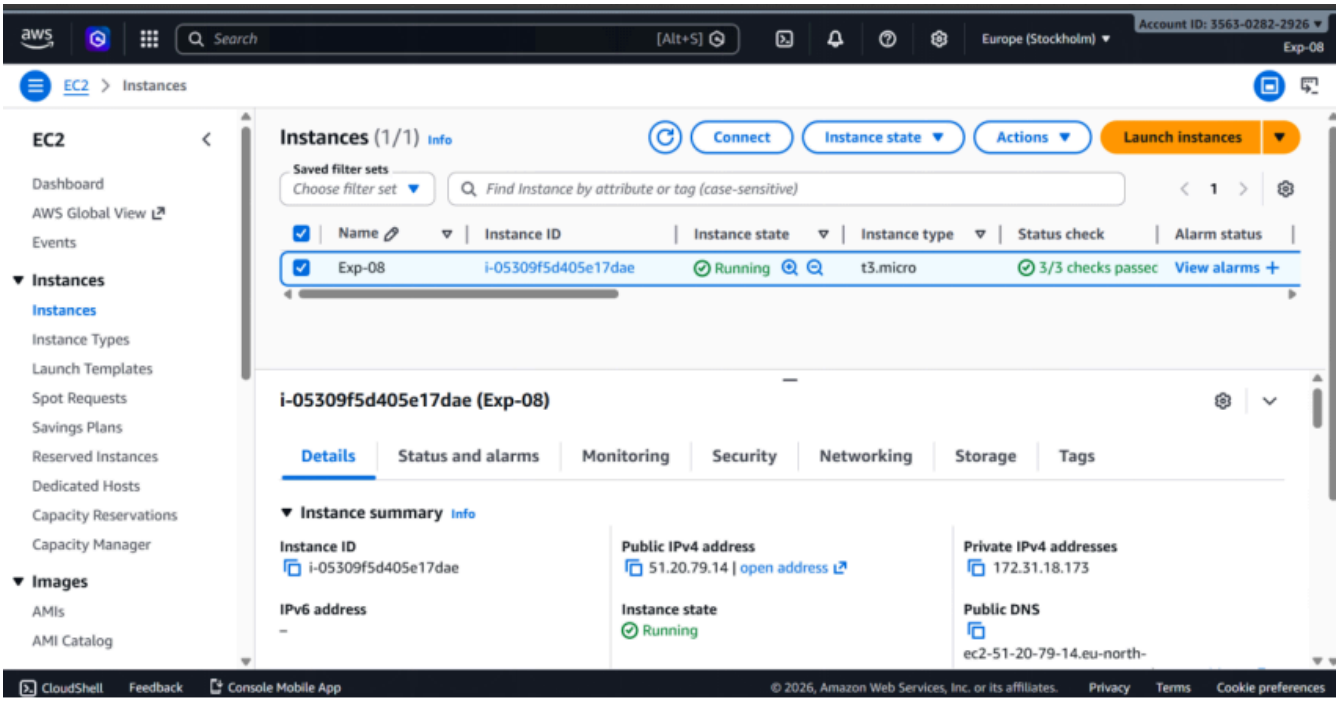



AWS Account
Account ID
356302822926
Account Alias
exp-08 Edit | Delete
Sign-in URL for IAM users in this account
https://exp-08.signin.aws.amazon.com/console


User details
User name
Exp-08
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)
Provide user access to the AWS Management Console - optional
In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.
Console password
Autogenerated password
You can view the password after you create the user.
Custom password
Enter a custom password for the user.
Show password
Users must create a new password at next sign-in - Recommended
Users automatically get the IAMUserChangePassword policy to allow them to change their own password.
If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more
Cancel
Next


download this password.
Console sign-in details
Email sign-in instructions
Console sign-in URL
https://exp-08.signin.aws.amazon.com/console
User name
Exp-08
Console password
********** Show
Cancel
Download .csv file
Return to users list





Conclusion:

Implementation of IAM (Identity and Access Management) ensures secure and controlled access to cloud resources by defining who can access what and under which conditions. It involves creating users, roles, and policies, applying the principle of least privilege, enabling multi-factor authentication (MFA), and continuously monitoring access activities. Proper IAM implementation enhances security, prevents unauthorized access, and helps organizations maintain compliance while efficiently managing permissions across systems.